Space physics

30 questions: 15 multiple choice, 15 structured.

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1 · Part 1 Core The Earth [1]

Each day the Sun appears to rise in the east, move across the sky and set in the west.

Which statement explains this observation?

A The Sun orbits the Earth once every 24 hours.

B The Earth orbits the Sun once every 24 hours.

C The Earth rotates on its axis once in approximately 24 hours.

D The axis of the Earth is tilted.

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2 · Part 1 Core The Earth [1]

A student sees a full Moon tonight.

Approximately how long will it be until the student next sees a full Moon?

A one month

B one week

C one day

D one year

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3 · Part 1 Core The Solar System [1]

Which list gives four planets in order of increasing distance from the Sun?

A Earth, Mars, Saturn, Jupiter

B Mercury, Earth, Venus, Mars

C Jupiter, Saturn, Neptune, Uranus

D Venus, Mars, Jupiter, Uranus

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4 · Part 1 Core The Solar System [1]

Which row correctly compares the planet Mars with the planet Neptune?

	Mars	Neptune
A	gaseous and large	rocky and small
B	rocky and small	gaseous and large
C	rocky and large	gaseous and small
D	gaseous and small	rocky and large

5 · Part 1 Core The Solar System [1]

What provides the force that keeps the Earth in its orbit around the Sun?

A the gravitational attraction of the Sun on the Earth

B the gravitational attraction of the Moon on the Earth

C the magnetic attraction between the Sun and the Earth

D the rotation of the Earth on its axis

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6 · Part 1 Extended The Solar System [1]

Planet X orbits the Sun at a greater distance than planet Y.

Which statement is correct?

A At planet X, the Sun's gravitational field is stronger, and planet X has a greater orbital speed than planet Y.

B At planet X, the Sun's gravitational field is stronger, and planet X has a smaller orbital speed than planet Y.

C At planet X, the Sun's gravitational field is weaker, and planet X has a smaller orbital speed than planet Y.

D At planet X, the Sun's gravitational field is weaker, and planet X has a greater orbital speed than planet Y.

7 · Part 1 Extended The Solar System [1]

A comet moves around the Sun in a long, narrow elliptical orbit.

Where is the Sun?

A at the centre of the ellipse

B inside the ellipse but not at its centre, nearer to one end of the orbit

C on the path of the orbit, at one end of the ellipse

D outside the ellipse

8 · Part 1 Core The Sun as a star [1]

In which regions of the electromagnetic spectrum does the Sun radiate most of its energy?

A gamma rays, X-rays and ultraviolet

B radio waves, microwaves and infrared

C visible light, X-rays and gamma rays

D infrared, visible light and ultraviolet

9 · Part 1 Core Stars [1]

What is a light-year?

A the distance travelled by light in space in one year

B the time taken for light to travel from the Sun to the Earth

C the length of one year measured on a planet orbiting a distant star

D the speed of light measured over a period of one year

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10 · Part 1 Extended Stars [1]

A star with a much greater mass than the Sun runs out of hydrogen in its centre.

Which sequence describes what happens to this star next?

A red giant → planetary nebula → white dwarf

B red supergiant → planetary nebula → white dwarf

C red supergiant → supernova → neutron star or black hole

D red giant → supernova → white dwarf

11 · Part 1 Core The Universe [1]

Which statement describes redshift?

A a decrease in the observed wavelength of light from stars that are moving towards the Earth

B an increase in the observed wavelength of electromagnetic radiation from stars and galaxies that are moving away

C an increase in the observed frequency of light from galaxies that are moving away

D the change in colour of a star as it cools and becomes a red giant

12 · Part 1 Core The Universe [1]

Astronomers find that the light from distant galaxies is redshifted compared with light emitted on the Earth.

What does this observation provide evidence for?

A The distant galaxies are moving towards the Earth.

B The distant galaxies contain only cool red stars.

C The Universe is getting smaller.

D The Universe is expanding, which supports the Big Bang Theory.

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13 · Part 1 Extended The Universe [1]

Which statement about the cosmic microwave background radiation (CMBR) is correct?

A. It is microwave radiation of a specific frequency that is observed at all points in space around us.

B. It is microwave radiation that is emitted by the surface of the Sun.

C. It is microwave radiation that is observed only from the direction of the centre of the Milky Way.

D. It is infrared radiation from nearby stars that has been reflected by dust in space.

14 · Part 1 Extended The Universe [1]

A galaxy is 4.0×10^{24} m from the Earth.

Use the current estimate of the Hubble constant to find the speed at which this galaxy is moving away from the Earth.

A. 5.5×10^{-43} m / s

B. 8.8×10^{6} m / s

C. 8.8×10^{7} m / s

D. 1.8×10^{42} m / s

15 · Part 1 Extended The Universe [1]

How can astronomers determine the distance d of a far galaxy?

A. by sending a radar signal to the galaxy and timing the echo

B. by measuring the mass of the galaxy

C. by measuring the brightness of a supernova in the galaxy

D. by measuring the frequency of the microwave radiation arriving from the direction of the galaxy

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16 · Part 1 Core The Earth [5]

A student keeps a diary of sunrise and sunset times in her city for a whole year.

(a) State the approximate time taken for the Earth to rotate once on its axis. [1]

(b) Explain why the city has a periodic cycle of day and night. [2]

(c) The student notices that summer days are longer and warmer than winter days, and that this pattern repeats every year.

Explain why the seasons repeat in this way. [2]

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17 · Part 1 Extended The Earth [6]

A large planet has several moons. One of its moons moves in an approximately circular orbit of average radius 6.7×10^8 m. The orbital period of this moon is 3.55 days.

(a) State the equation that defines the average orbital speed v of an object in an orbit of average radius r and orbital period T. [1]

(b) Calculate the average orbital speed of the moon. [3]

(c) A space probe orbits the same planet in a circular orbit of radius 5.4×10^8 m at an average orbital speed of 1.5×10^4 m / s.

Calculate the orbital period of the space probe in days. [2]

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18 · Part 1 Core The Solar System [5]

The speed of light in space is 3.0×10^8 m / s. Radio waves travel through space at the same speed as light.

(a) The average distance between the Sun and Mars is 2.3×10^{11} m.

Calculate the time taken for light from the Sun to reach Mars. Give your answer in minutes. [3]

(b) A rover on the surface of Mars sends a radio message to a control centre on the Earth. At this time, the distance between the Earth and Mars is 7.8×10^{10} m. The control centre sends a reply as soon as it receives the message.

Calculate the shortest time between the rover sending its message and the rover receiving the reply. [2]

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19 · Part 1 Core The Solar System [6]

Venus is one of the four planets nearest the Sun. Saturn is one of the four planets furthest from the Sun.

(a) Compare Venus with Saturn in terms of their size and what they are made of. [2]

(b) Scientists use an accretion model to explain how the Solar System formed.

Explain how the Sun and the planets formed, and why the planets nearest the Sun are different from the planets furthest from the Sun. [4]

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20 · Part 1 Core The Solar System [5]

(a) State what the strength of the gravitational field at the surface of a planet depends on. [1]

(b) Jupiter has a much greater mass than the Earth.

State how the gravitational field strength at the surface of Jupiter compares with the gravitational field strength at the surface of the Earth. [1]

(c) A space probe travels away from the Earth into deep space.

Describe how the strength of the Earth's gravitational field at the position of the probe changes during this journey. [1]

(d) Explain why all the planets in the Solar System orbit the Sun. [2]

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21 · Part 1 Extended The Solar System [7]

The table gives data about six planets.

planet	average orbital distance from the Sun / 10^6 km	orbital duration / Earth years	average surface temperature / °C	gravitational field strength at surface / N / kg
Mercury	58	0.24	167	3.7
Venus	108	0.62	464	8.9
Earth	150	1.00	15	9.8
Mars	228	1.88	−65	3.7
Jupiter	778	11.9	−110	24.8
Saturn	1430	29.5	−140	10.4

(a) Describe the relationship between the average orbital distance of a planet and its orbital duration. [1]

(b) Describe the general trend in average surface temperature as the orbital distance increases, and identify the planet that does not follow this trend. [2]

(c) Assume the orbit of the Earth is circular. Use the data in the table to calculate the average orbital speed of the Earth in m / s. [3]

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(d) Suggest why the gravitational field strength at the surface of Jupiter is much greater than at the surface of the Earth. [1]

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22 · Part 1 Extended The Solar System [5]

A comet travels around the Sun in an elliptical orbit. Point P is the point on the orbit closest to the Sun. Point Q is the point on the orbit furthest from the Sun.

(a) State whether the comet is travelling fastest at point P or at point Q. [1]

(b) Explain, using the conservation of energy, why the speed of the comet changes as it moves from Q to P. [3]

(c) A planet moves around the Sun in an orbit that is almost circular.

Explain why the speed of the planet stays almost constant. [1]

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23 · Part 1 Extended The Sun as a star [5]

The Sun is a stable star of medium size.

(a) State the two elements that make up most of the Sun. [1]

(b) Energy is released in the Sun by nuclear reactions.

State the type of nuclear reaction that takes place in a stable star, and describe what happens in this reaction. [2]

(c) Explain why the Sun is stable and neither collapses nor expands. [2]

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24 · Part 1 Extended Stars [7]

Star A has a mass similar to the mass of the Sun. Star B has a much greater mass than the Sun.

(a) Both stars formed from interstellar clouds of gas and dust.

Describe how a protostar forms from an interstellar cloud. [2]

(b) Describe what happens to star A after most of the hydrogen in its centre has been converted to helium. [2]

(c) Describe what happens to star B after most of the hydrogen in its centre has been converted to helium. [2]

(d) Star B is 640 light-years from the Earth.

Calculate this distance in metres. [1]

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25 · Part 1 Core The Universe [4]

The Sun is one of many billions of stars in a galaxy.

(a) State the name of the galaxy that contains the Sun. [1]

(b) State the approximate diameter of this galaxy in light-years. [1]

(c) State approximately how many galaxies make up the Universe. [1]

(d) Compare the distance from the Earth to the other stars in this galaxy with the distance from the Earth to the Sun. [1]

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26 · Part 1 Core The Universe [5]

In a laboratory on the Earth, hot hydrogen gas gives out red light with a wavelength of 6.56×10^{-7} m.

An astronomer measures the same light from hydrogen in a distant galaxy. The observed wavelength is 6.89×10^{-7} m.

(a) Describe what is meant by redshift. [2]

(b) Calculate the increase in the wavelength of the light from the distant galaxy. [1]

(c) Light from almost all distant galaxies is redshifted.

Explain how this observation supports the Big Bang Theory. [2]

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27 · Part 1 Extended The Universe [4]

Radio telescopes detect cosmic microwave background radiation (CMBR).

(a) State one feature of the CMBR which shows that it does not come from a single object, such as one star or one galaxy. [1]

(b) Explain when the CMBR was produced and why it is now detected in the microwave region of the electromagnetic spectrum. [3]

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28 · Part 1 Extended The Universe [7]

An astronomer studies a far galaxy J. Galaxy J is 3.0×10^{24} m from the Earth and is moving away from the Earth at a speed of 6.9×10^{6} m / s.

(a) State how the speed at which a galaxy is moving away from the Earth can be found. [1]

(b) Define the Hubble constant H0. [1]

(c) Calculate the value of the Hubble constant from the data for galaxy J. [2]

(d) Another galaxy K is moving away from the Earth at a speed of 4.4×10^{6} m / s.

Use the current estimate of the Hubble constant to calculate the distance of galaxy K from the Earth. Give your answer in metres and in light-years. [3]

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29 · Part 1 Extended The Universe [6]

The Hubble constant can be used to estimate the age of the Universe.

(a) State the current estimate for the Hubble constant H0. Include its unit. [1]

(b) Calculate an estimate for the age of the Universe in seconds. [2]

(c) Calculate the age of the Universe in years. [2]

(d) Explain why this calculation is evidence that all the matter in the Universe was once present at a single point. [1]

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30 · Part 1 Extended The Universe [6]

Astronomers observe a supernova in a far galaxy. They use the supernova to find that the galaxy is 1.2×10^{25} m from the Earth.

(a) State which property of the supernova is measured to determine the distance of the galaxy. [1]

(b) State what is measured in the light from the galaxy to find the speed at which the galaxy is moving away from the Earth. [1]

(c) Assume that the distance between the galaxy and the Earth has not changed while the light was travelling.

Calculate how many years ago the supernova exploded. [3]

(d) Suggest why astronomers use supernovae, and not ordinary stars, to determine the distances of far galaxies. [1]

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